

Networks, Exchange Rates, and Monetary Policy in a Small Open Economy

Extending Rubbo (2024) to FX Regime Analysis

[Your Name]

[Your Institution]

July 22, 2026

Outline

- 1 Motivation
- 2 Literature Review
- 3 Closed-Economy Model: Rubbo (2024)
- 4 Small Open Economy Extension
- 5 Data and Quantitative Strategy
- 6 Preliminary Results
- 7 Conclusion

Motivation: Two Puzzles

Puzzle 1: Which Inflation to Target?

- Multi-sector economies: CPI includes many sectors with different stickiness
- Rubbo (2024): optimal target is *not* CPI — it is a network-adjusted index
- **Q: How does this change in an open economy?**

Puzzle 2: Float vs. Peg?

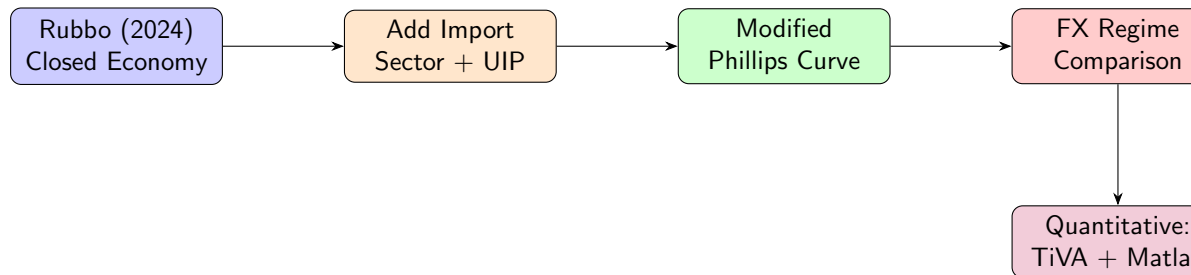
- Exchange rate movements propagate through supply chains
- Sectors with upstream import linkages amplify pass-through
- **Q: What FX regime minimizes welfare loss in a networked economy?**

This Paper

We extend Rubbo (2024) to a small open economy and show:

- 1 The “divine coincidence” breaks down under a float (two cost-push channels)
- 2 Network structure determines optimal FX policy
- 3 Countries with heterogeneous import intensity benefit most from FX intervention

Roadmap



Key building blocks from Rubbo (2024):

- Domar weights, Leontief inverse, rigidity-adjusted network
- Divine coincidence index, sector-level Phillips curves
- Welfare: output gap + within-sector + cross-sector misallocation

Literature: Three Intersecting Areas

Area 1: Networks & MP

- Rubbo (2024, Ecta): sector Phillips curves, DC index
- La'O & Tahbaz-Salehi (2022, Ecta): optimal MP in networks
- Pasten-Schoenle-Weber (2020, JME): heterogeneous Calvo + networks
- Baqaee & Farhi (2020, QJE): misallocation in networks
- Acemoglu et al. (2012, Ecta): micro shocks → macro

Area 2: Intl. Networks + ER

- Auer-Levchenko-Sauré (2019, REStat): IO spillovers of inflation
- Amiti-Itskhoki-Konings (2019, ReStud): markups + pass-through
- Gopinath & Itskhoki (2022): dominant currency paradigm
- Bems & Johnson (2017, AEJMacro): value-added REERs
- GVC literature: networks dampen pass-through

Area 3: FX Policy

- Galí & Monacelli (2005, ReStud): baseline SOE NK model
- Corsetti-Dedola-Leduc (2010): optimal MP in open economies
- Fanelli & Straub (2021, ReStud): FX intervention theory
- Schmitt-Grohé & Uribe (2016, JPE): peg + wage rigidity
- Qiu et al. (2026, JME): MP + production networks in open econ

Our Contribution vs. Existing Literature

	Networks	Open Econ.	FX Policy
Rubbo (2024)	✓	×	×
La'O & Tahbaz-Salehi (2022)	✓	×	×
Galí & Monacelli (2005)	×	✓	✓
Fanelli & Straub (2021)	×	✓	✓
Qiu et al. (2026)	✓	✓	×
Auer et al. (2019)	✓	✓	×
This paper	✓	✓	✓

Key gap we fill

No paper combines: (i) full network structure with sector-level Phillips curves, (ii) a small open economy, *and* (iii) an analysis of FX policy regimes in terms of the divine coincidence/welfare.

Model: Production Network

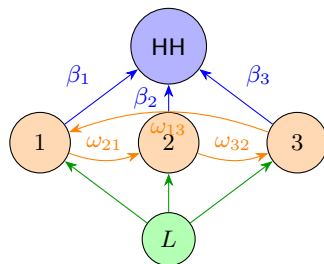
N **sectors**, each a continuum of Calvo firms:

$$Y_{if} = A_i F_i(L_{if}, \{X_{ijf}\}) \quad (\text{CRS})$$

$$P_i = \left(\int P_{if}^{1-\varepsilon_i} df \right)^{1/(1-\varepsilon_i)}$$

Network notation:

- ω_{ij} = cost share of input j in sector i (IO matrix Ω)
- α_i = labor cost share in sector i
- β_i = household spending share on sector i
- $\hat{\delta}_i$ = effective Calvo frequency (price flexibility)



Key Objects: Leontief Inverse and Domar Weights

Log-linearized Marginal Cost

$$mc_t = \underbrace{\alpha w_t}_{\text{labor cost}} + \underbrace{\Omega p_t}_{\text{input cost}} - \underbrace{\log \mathbf{A}_t}_{\text{TFP}}$$

Leontief Inverse

$$(I - \Omega)^{-1} = I + \Omega + \Omega^2 + \dots$$

$(I - \Omega)^{-1}_{ij}$ = total exposure of i 's cost to j 's price, through *all* supply chain paths.

Domar Weights

$$\lambda^T = \beta^T (I - \Omega)^{-1}$$

λ_i = sector i 's total sales share (direct to consumers + indirect through other producers). Always $\sum_i \lambda_i > 1$.

Rigidity-Adjusted Leontief $(I - \Omega\Delta)^{-1}$

Weights each step of the supply chain by sector's price flexibility $\hat{\delta}_i$. Sticky-price sectors absorb cost shocks into markups rather than prices — they are “discounted” in propagation.^{8/30}

Three Lemmas (Rubbo 2024)

Lemma 1: Natural Output = Domar-Weighted TFP

$$y_{t,\text{nat}} = \frac{1 + \varphi}{\gamma + \varphi} \boldsymbol{\lambda}^T \log \mathbf{A}_t$$

Natural output is a weighted sum of TFPs; weights = Domar weights (network centrality).

Lemma 2: Output Gap = Employment Gap

$$\tilde{y}_t = y_t - y_{t,\text{nat}} = l_t - l_{t,\text{nat}} = \tilde{l}_t$$

Because labor is allocated efficiently *given* aggregate L_t (envelope theorem).

Lemma 3: Output Gap = -Sales-Weighted Markup

$$(\gamma + \varphi)\tilde{y}_t = -\boldsymbol{\lambda}^T \boldsymbol{\mu}_t = -\sum_i \lambda_i \mu_{it}$$

Positive markups distort labor supply. Effect is amplified for sectors with high Domar weight.

Proposition 2: Sector-Level Phillips Curves

$$\pi_t = \underbrace{\rho(I - \mathcal{V})\mathbb{E}\pi_{t+1}}_{\text{forward-looking}} + \underbrace{\mathcal{B}(\gamma + \varphi)\tilde{y}_t}_{\text{output gap}} - \underbrace{\mathcal{V}\chi_t}_{\text{cost-push}}$$

- $\mathcal{B} = \frac{\Delta(I - \Omega\Delta)^{-1}\alpha}{1 - \beta^T\Delta(I - \Omega\Delta)^{-1}\alpha}$
Phillips curve slope (network-adjusted labor share)
- $\mathcal{V}$: cost-push matrix (rows sum to 0)
Only *relative* productivity matters
- $(I - \Omega\Delta)^{-1}$: rigidity-adjusted Leontief
Sticky sectors dampen cost propagation
- $\chi_t = (I - \Omega)^{-1} \log \mathbf{A}_t - \mathbf{p}_{t-1}$
Cost-push = efficiency-adjusted price wedge

Key: Sherman-Morrison identity

The complex matrix inversion in the proof uses: $(I - \mathbf{u}\mathbf{v}^T)^{-1} = I + \frac{\mathbf{u}\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}}$, allowing closed-form solution for $\mathcal{B}$ and $\mathcal{V}$.

Proposition 1: Divine Coincidence Index

Definition

$$\pi_t^{DC} \equiv \sum_i \frac{\lambda_i(1 - \hat{\delta}_i)/\hat{\delta}_i}{\underbrace{\sum_j \lambda_j(1 - \hat{\delta}_j)/\hat{\delta}_j}_{= w_i^{DC}}} \pi_{it}$$

Weight $\propto$ Domar weight $\times$ price stickiness. Stickier sectors get *more* weight.

The DC Phillips Curve: No Cost-Push!

$$\pi_t^{DC} = \rho \mathbb{E} \pi_{t+1}^{DC} + \frac{\gamma + \varphi}{\underbrace{\sum_j \lambda_j(1 - \hat{\delta}_j)/\hat{\delta}_j}_{\kappa_{DC}}} \tilde{y}_t$$

Uniqueness: π^{DC} is the *unique* inflation index with no cost-push term (one-dimensional left null space of $\mathcal{V}$).

Policy implication: Replace CPI with π^{DC} in the Taylor rule $\Rightarrow$ zero output gap at all times.

Welfare (Proposition 3)

$$\mathbb{W} = \frac{1}{2} \sum_t \rho^t \left[\underbrace{(\gamma + \varphi) \tilde{y}_t^2}_{(1) \text{ output gap}} + \underbrace{\sum_i \lambda_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \varepsilon_i \pi_{it}^2}_{(2) \text{ within-sector}} + \underbrace{\Phi_C + \sum_s \lambda_s \Phi_s}_{(3) \text{ cross-sector}} \right]$$

- ① **Output gap:** Same as one-sector NK model. Labor-leisure distortion.
- ② **Within-sector price dispersion:** Under Calvo, variance of prices within sector $i \approx (1 - \hat{\delta}_i) \pi_{it}^2 / \hat{\delta}_i$. Consumers buy from cheap (not productive) varieties.
- ③ **Cross-sector misallocation:** Relative price wedges cause inefficient substitution across sectors. Captured by Baqaee-Farhi substitution operators Φ .

Key implication

Even if output gap = 0, the CB should care about *which* sectors' prices inflate, because sectors differ in stickiness and network position.

Open Economy: Modified Environment

Add foreign sector F :

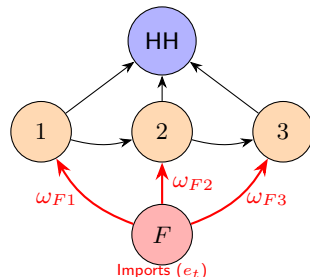
- Consumption basket: $C_t = \mathcal{C}(C_t^H, C_t^F)$
- CPI: $p_t^C = \mathcal{P}(p_t^H, e_t + p_t^*)$
- $e_t = \log$ nominal exchange rate (domestic/foreign)
- $p_t^* =$ foreign price level (exogenous)

Domestic firms use imports:

$$mc_t = \alpha w_t + \Omega p_t + \underbrace{\omega_F(p_t^* + e_t)}_{\text{new term}} - \log \mathbf{A}_t$$

ω_{Fi} = import input share of sector i (from TiVA data).

Uncovered Interest Parity (UIP): $i_t - i_t^* = \mathbb{E}_t \Delta e_{t+1}$



Modified Phillips Curve

Phillips Curve with Exchange Rate Pass-Through

$$\pi_t = \rho(I - \mathcal{V})\mathbb{E}\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \underbrace{\Gamma\Delta e_t}_{\text{new}}$$

Exchange Rate Pass-Through Vector

$$\Gamma \equiv \frac{\Delta(I - \Omega\Delta)^{-1}\omega_F}{1 - \beta^T\Delta(I - \Omega\Delta)^{-1}\alpha}$$

Interpretation of Γ_i :

- Direct effect: sector i 's own import intensity ω_{Fi}
- **Network amplification:** $(I - \Omega\Delta)^{-1}$ adds indirect effects through upstream suppliers' import intensity
- Sectors with stickier prices ($\hat{\delta}_i$ small) absorb more into markups: smaller Γ_i
- Sectors with many import-intensive upstream suppliers: larger Γ_i

Network Amplification of FX Pass-Through

Expand the rigidity-adjusted Leontief:

$$(I - \Omega\Delta)^{-1}\omega_F = \omega_F + \Omega\Delta\omega_F + (\Omega\Delta)^2\omega_F + \dots$$

Term	Interpretation
ω_F	Direct import intensity of each sector
$\Omega\Delta\omega_F$	Import intensity of direct suppliers, weighted by their price flexibility
$(\Omega\Delta)^2\omega_F$	Import intensity of suppliers' suppliers
$\vdots$	$\dots$

Key Testable Prediction

Sectors with high **upstream** import intensity (even if their own ω_{Fi} is low) experience large inflation responses to exchange rate movements. This is the network amplification effect, measurable from TiVA data.

Result 1: Breakdown of Divine Coincidence

Under a float: we need a weighting ϕ such that $\phi^T \pi_t$ has no cost-push. Two conditions must hold simultaneously:

$$\underbrace{\phi^T \mathcal{V} = \mathbf{0}^T}_{\text{no productivity cost-push}} \quad \text{AND} \quad \underbrace{\phi^T \Gamma = 0}_{\text{no ER cost-push}}$$

Closed economy (Rubbo)

One condition $\Rightarrow$ unique solution: the DC index ϕ^{DC} . **Divine coincidence holds.**

Open economy (this paper)

Two conditions, one free parameter.

Generically no solution — unless $\Gamma \propto \mathcal{B}$ (ER affects sectors exactly like wages do).

Divine coincidence breaks down.

Proposition (informal)

Under a flexible exchange rate, there is an irreducible tradeoff between output gap stability and inflation stability. The severity of the tradeoff is governed by $|\Gamma - \mathcal{B} \cdot c|$ for any scalar c .

FX Regime Comparison

Property	Float	Peg	Managed Float
ER cost-push $\Gamma \Delta e_t$	Yes	Zero	Reduced
MP independence	Full	None	Partial
DC index survives?	No	Yes (trivially)	Depends on ν_t
Optimal inflation target	No single index	DC (can't use it)	Modified DC
Welfare			
Output gap loss	Low (can target)	High (trilemma)	Intermediate
ER misallocation	High	Zero	Low
Total	?	?	Best?

Trilemma (Mundell)

Under a peg: $\Delta e_t = 0$ eliminates ER cost-push. But $i_t = i_t^*$ by UIP: CB loses monetary policy independence. Cannot close output gap. Cross-sector misallocation from foreign shocks becomes the main welfare cost.

Result 2: Optimal FX Intervention

Under a managed float: CB has two instruments — interest rate i_t + FX intervention ν_t .

Optimal Policy

- ① Use i_t to close the output gap ($\tilde{y}_t = 0$)
- ② Use ν_t to offset the ER cost-push component that is orthogonal to $\mathcal{B}$:

$$\nu_t^* = -\frac{\phi^{DC,T}\Gamma}{\phi^{DC,T}\Gamma_\nu} \Delta e_t^{\text{unmanaged}}$$

where Γ_ν is the pass-through of FX intervention to exchange rate

Intuition

The CB should “lean against” exchange rate movements in proportion to their network-amplified pass-through. The optimal intervention depends on Γ — which depends on the country’s IO structure and import intensity distribution ω_F .

Modified Welfare in the Open Economy

$$\mathbb{W}^{\text{open}} = \underbrace{\mathbb{W}^{\text{closed}}}_{\text{Rubbo (2024)}} + \underbrace{\frac{1}{2} \sum_t \rho^t \left[\Phi_C(\tilde{\chi}_t, e_t \omega_F) + \sum_s \lambda_s \Phi_s(\tilde{\chi}_t, e_t \omega_F) \right]}_{\text{ER-driven cross-sector misallocation (new)}}$$

New term: Exchange rate movements cause relative price distortions across sectors with different import intensities ω_{Fi} . When the ER depreciates, import-intensive sectors face higher costs than others; relative prices diverge from efficient levels.

Counterfactual: What Would CPI Targeting Do?

- CPI targeting ignores within-sector dispersion and cross-sector misallocation
- Under a float + CPI targeting: both output gap and ER misallocation accumulate
- Rubbo shows: welfare loss of 2.9–3.8% of GDP vs. optimal even in *closed* economy
- Our conjecture: losses are larger under a float, especially for countries with high $\|\Gamma\|$

Data: OECD TiVA and Country Selection

OECD Trade in Value Added (TiVA):

- Inter-country IO tables, 66 countries, 45 sectors
- Covers: Ω (domestic IO), ω_F (import shares by sector), β (consumption shares), λ (Domar weights)
- Years: 2005, 2010, 2015, 2018 (latest)

Country selection criteria:

- Small open economy (GDP < 1% of world)
- Heterogeneous FX regimes (float, peg, managed)
- Rich IO data
- Candidate: South Korea, Chile, Czech Republic, New Zealand, Singapore

Key Parameters to Calibrate

- Ω : from TiVA domestic IO table
- ω_F : import use by sector (TiVA)
- β : final consumption shares (TiVA)
- $\hat{\delta}$: sectoral Calvo frequencies from Nakamura-Steinsson (2008) mapped to TiVA sectors
- ε_i : substitution elasticities from literature
- $\gamma = 1, \varphi = 2$: standard NK calibration
- $\rho = 0.9975$: quarterly discount factor

Quantitative Objects of Interest

1. Pass-through vector Γ (theory $\rightarrow$ data):

$$\Gamma = \frac{\Delta(I - \Omega\Delta)^{-1}\omega_F}{1 - \beta^T \Delta(I - \Omega\Delta)^{-1}\alpha}$$

Compare direct import intensity ω_F vs. network-amplified Γ : how much does the network matter?

2. DC index weights (country comparison):

$$w_i^{DC} = \frac{\lambda_i(1 - \hat{\delta}_i)/\hat{\delta}_i}{\sum_j \lambda_j(1 - \hat{\delta}_j)/\hat{\delta}_j}$$

How much do DC weights differ from CPI weights? How much do they differ across countries?

3. Welfare comparison across FX regimes (simulation):

- Simulate model under Float / Peg / Managed Float
- Compute $\mathbb{W}^{\text{open}}$ for each regime, for each country
- Identify: which countries should float? Which should peg?

Matlab Simulation Plan

Extends Rubbo's existing code (replication files/calibration/):

Task	File	Change from Rubbo
Compute $\Omega, \alpha, \beta, \hat{\delta}$	IO_clean.m, deltas.m	Add ω_F from TiVA import col.
Compute $\Gamma, \mathcal{B}, \mathcal{V}$	parameters_d.m	Add $\text{Gamma} = \text{Delta} * (\text{I} - \text{Omega} * \text{Delta})^{(-1)} * \text{omega}$
Solve RE equilibrium	pol_fct.m.m	Add Δe_t to state vector
Float IRFs	IRFs_monetary.m	Add UIP equation
Peg IRFs	New IRFs_peg.m	Fix $\Delta e_t = 0, i_t = i_t^*$
Welfare comparison	simul.m + new	Compute $\mathbb{W}^{\text{open}}$ for Float/Peg/Managed

Key addition to Matlab code

```
omega_F = tiva_import_shares; %  $N \times 1$ , from TiVA data
Gamma = Delta*(eye(n)-Omega*Delta)^(-1)*omega_F / den;
% Add  $\Gamma \Delta e_t$  to T matrix;  $\Delta e_t$  from UIP as new state variable
```

Preliminary Theoretical Results

Result 1: Network Amplification of Pass-Through

$\Gamma \geq \omega_F$ element-wise (weakly). The network always amplifies direct import intensity. The amplification is stronger when:

- Upstream suppliers are import-intensive (ω_{Fj} large for j supplying i)
- Upstream suppliers have flexible prices ($\hat{\delta}_j$ large — they pass through more)

Result 2: DC Breakdown Under Float

The DC index fails to eliminate cost-push iff $\Gamma \notin \text{span}(\mathcal{B})$. Since $\mathcal{B} \propto (I - \Omega\Delta)^{-1}\alpha$ and $\Gamma \propto (I - \Omega\Delta)^{-1}\omega_F$, the condition for DC survival is $\omega_F \propto \alpha$ (import intensity proportional to labor intensity). This fails for any realistic IO structure.

Result 3: Peg as DC Restorer

Under a peg, $\Delta e_t = 0$ eliminates the $\Gamma\Delta e_t$ term. The closed-economy DC index is again the relevant inflation target. But the Mundell trilemma prevents the CB from using it.

Expected Quantitative Findings (Preliminary)

Based on calibration to Korea (2018 TiVA):

- $\|\Gamma - \omega_F\|/\|\omega_F\| \approx [25\text{--}40\%]$
(network amplifies direct pass-through by $\sim 30\%$)
- DC index vs. CPI: sticky-price sectors (services) get higher DC weight; flexible-price sectors (food, energy) get lower weight
- Welfare under float: $[X\%]$ of GDP additional loss vs. optimal, mainly from ER misallocation
- Optimal FX intervention offsets $\sim [Y\%]$ of welfare loss vs. pure float

Country comparison (expected):

- Countries with high $\|\Gamma\|$ (e.g., Singapore: very open, high import intensity): large gains from managed float
- Countries with homogeneous ω_{Fi} (e.g., New Zealand: commodity-dominated): smaller gains, $\Gamma \approx c \cdot \mathcal{B}$, DC near-survives
- Countries with heterogeneous import intensity across sectors: DC breakdown is severe; managed float strongly preferred

Summary and Contribution

Model

Multi-sector NK model with IO networks, extended to a SOE:

- N domestic sectors + import sector; Calvo pricing; UIP; LOP
- Exchange rate pass-through through production chains: $\Gamma = \Delta(I - \Omega\Delta)^{-1}\omega_F/\kappa$

Theoretical Contributions

- 1 **Network amplification:** ER pass-through amplified by upstream import linkages
- 2 **DC breakdown:** No single inflation index eliminates cost-push under a float
- 3 **Optimal FX:** CB should intervene proportional to Γ to restore tractability
- 4 **Country heterogeneity:** Countries with dispersed import intensity benefit most from intervention

Quantitative

OECD TiVA data, Korea/Chile/Czech Republic/NZ calibration, Matlab simulations building on Rubbo's codebase.

Next Steps and Timeline

Task	Status	Target
Proof of Results 1–3	In progress	June 30
TiVA data download + ω_F	Not started	July 1
Matlab: add Γ to codebase	Not started	July 5
IRFs under float/peg/managed	Not started	July 10
Welfare comparison tables	Not started	July 14
Draft Sections 2–4 (paper)	Not started	July 18
Presentation	This draft	July 22

Open questions for discussion

- 1 Should we allow variable markups / pricing-to-market? (Amiti-Itskhoki-Konings 2019)
- 2 How to handle dominant currency pricing? (Gopinath 2022)
- 3 Should we target one country or do a panel comparison?

References (Selected)

- Rubbo, E. (2024). *Networks, Phillips Curves, and Monetary Policy*. *Econometrica* 91(3).
- La'O, J. & Tahbaz-Salehi, A. (2022). *Optimal Monetary Policy in Production Networks*. *Econometrica* 90(3).
- Pasten, E., Schoenle, R. & Weber, M. (2020). *Propagation of MP Shocks in a Heterogeneous Production Economy*. *JME* 116.
- Auer, R., Levchenko, A. & Sauré, P. (2019). *International Inflation Spillovers Through Input Linkages*. *REStat* 101(3).
- Amiti, M., Itskhoki, O. & Konings, J. (2019). *International Shocks, Variable Markups, Domestic Prices*. *ReStud* 86(6).
- Fanelli, S. & Straub, L. (2021). *A Theory of Foreign Exchange Interventions*. *ReStud* 88(6).
- Galí, J. & Monacelli, T. (2005). *Monetary Policy and Exchange Rate Volatility in a SOE*. *ReStud* 72(3).
- Schmitt-Grohé, S. & Uribe, M. (2016). *Downward Nominal Wage Rigidity, Currency Pegs, and Involuntary Unemployment*. *JPE* 124(5).
- Qiu, Z. et al. (2026). *Monetary Policy in Open Economies with Production Networks*. *JME* (forthcoming)

Thank you!

Questions and comments welcome.

[email] | [website] | github.com/Sugarkhuu/rubbo_2024

Appendix: Sherman-Morrison Derivation (Full)

Goal: Invert $I - \Delta\Omega - \Delta\alpha\beta^T$.

Step 1: Factor: $I - \Delta\Omega - \Delta\alpha\beta^T = (I - \Delta\Omega)(I - \Delta(I - \Omega\Delta)^{-1}\alpha\beta^T)$.

Step 2: Use Sherman-Morrison on the rank-1 part:

$$(I - \mathbf{u}\beta^T)^{-1} = I + \frac{\mathbf{u}\beta^T}{1 - \beta^T\mathbf{u}}, \quad \mathbf{u} = \Delta(I - \Omega\Delta)^{-1}\alpha$$

Step 3: Combine:

$$(I - \Delta\Omega - \Delta\alpha\beta^T)^{-1} = \left(I + \frac{\mathbf{u}\beta^T}{\kappa} \right) (I - \Delta\Omega)^{-1}$$

where $\kappa = 1 - \beta^T\mathbf{u}$.

Step 4: Apply to get $\mathcal{B}$: LHS inverse $\cdot \Delta\alpha = \frac{\mathbf{u}}{\kappa}(\gamma + \varphi)$ (after using $\kappa + \beta^T\mathbf{u} = 1$).

Appendix: Matlab State-Space System

System from IRFs_monetary.m:

$$\underbrace{\begin{pmatrix} z_t \\ \pi_t \\ y_t \end{pmatrix}}_{n+n+1} = T^{-1} R m_t + T^{-1} T_{\text{lag}} \begin{pmatrix} z_{t-1} \\ \pi_{t-1} \\ y_{t-1} \end{pmatrix}$$

T matrix (rows = Phillips curves, IS, Taylor rule):

$$T = \begin{pmatrix} Z & I - M^{-1} & M^{-1} \Delta \mathcal{B} \\ -Z/\rho & M^{-1}/\rho & -M^{-1} \Delta \mathcal{B}/\rho \\ \mathbf{0} & \mathbf{0} & \zeta(\gamma + \varphi)/\gamma + 1 \end{pmatrix}$$

where $M = I + \Delta \mathcal{V}(I - \Omega)$ encodes the Phillips curve dynamics and $Z = I - M^{-1} \Delta \mathcal{B} \lambda^T (I - \Delta) \Delta^{-1} / (\gamma + \varphi)$ encodes the DC-targeting property. QZ decomposition solves for unique non-explosive solution gx , hx .